7.4 TYPES OF DC GENERATOR

DC generators are classified according to the manner in which field windings are connected. The process of giving DC voltage to the field winding for producing magnetic field is called field excitation. The generators are classified based on the exciation, as:

1. Separately Excited DC Generator

In separately excited generator, the exciation to the field winding is given by an external DC supply as shown in Fig 7.6.

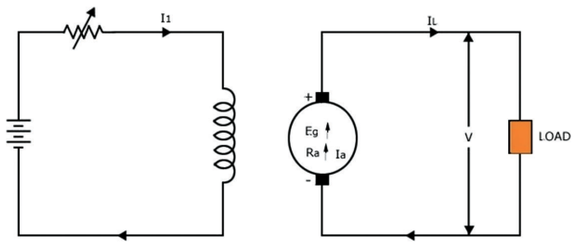


Fig 7.6 *Separately excited DC generator*

2. Self-Excited DC Generator

In this type of DC generators, the generation of emf is due to its property of residual flux or residual magnetism present in the field winding. They are mainly of three types:

a) DC series generator
b) DC shunt generator
c) DC compound generator

a) DC Series Generator

In DC series generators the field winding (R_{se}) is connected in series with the armature winding (R_a). Therefore, the series current (I_{se}) is equal to the armature current (I_a) and the load current (l_L), as shown in the Fig 7.7.

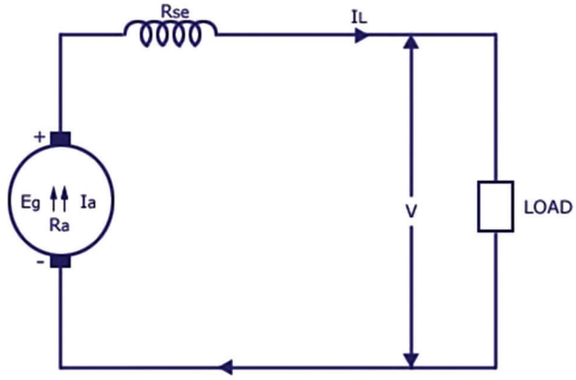


Fig 7.7 *DC Series Generator*

The current equation is $I_a = I_L = I_{se}$
The Voltage equation is $E_g = V + I_a(R_a + R_{se})$
V=load voltage

b) DC Shunt Generator

In shunt generator the field winding is across (shunt) the armature winding. Here the armature current (I_a) is the sum of shunt field current (I_{sh}) and load current (I_L) as shown in Fig 7.8.

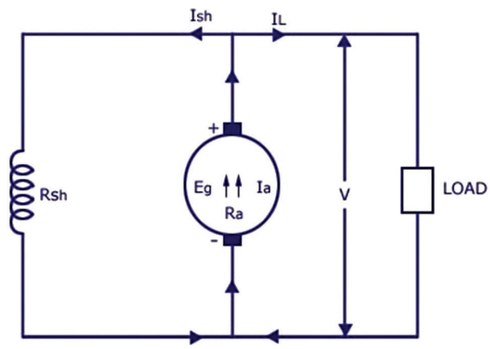


Fig 7.8 *DC Shunt Generator*

The current equation is $I_a = I_L + I_{sh}$
The voltage equation is $E_g = V + I_a R_a$

c) DC Compound Generator

These types of generators are again classified into long shunt generator and short shunt generator based on the field winding connections. In long shunt compound generator, the shunt field winding

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